



PROJECT-BASED LEARNING

PHASE-I EVALUATION

**GridRescue: Interactive Grid
Failure & Load Redistribution Simulator**

Team ID: T091

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T091

Semester: 3rd

Section: XX

Domain: *B.Tech 2nd-Year DSA+OOPS*

Team Members

1. *Tejasv Singh Nagarkoti –2510010772*
2. *Tanishq Vashisth –2510010163*
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4. *Riddhiman Sandilya –2510370168*

Mentor

Dr.Siddhant Thapliyal

Interactions completed:3

Problem Statement & Motivation

Problem Statement

A power grid has many connected components such as stations, substations and consumers. If one important component fails, the normal flow of power can get disturbed. Our project tries to simulate such failures on a small grid and show how the system can find another route and distribute the available power.

Motivation

- ▶ *To see how power grid works and how it behaves during a failure.*
- ▶ *when failure occurs we find the affected areas and possible alternate routes.*

Target Users / Stakeholders

- ▶ *Target Users: Power grid operators, electricity distribution teams.*
- ▶ *Stakeholders: Electricity companies, hospitals, etc.*

Objectives

Project Objectives

1. *Simulate real-world power grid operations and component failures.*
2. *Identify alternate power routes using Dijkstra's algorithm.*
3. *Prioritize power distribution using Min-Heaps and Priority Queues.*
4. *Apply OOP principles through modular C++ grid components.*
5. *Visualize grid failures and recovery through Dear ImGui.*

Key Objective: *To build an interactive system that helps detect grid failures, find alternate power routes, and redistribute power efficiently.*

Proposed Solution

Proposed Approach

1. *Set up a small 5-node power grid.*
2. *Connect the grid components using an adjacency list.*
3. *Use BFS, DFS and Dijkstra to find routes.*
4. *Use priority queues to handle important power loads.*
5. *Real-time graphical interface using Dear ImGui.*

Key Features

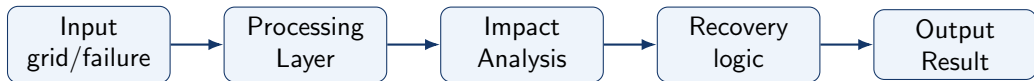
- ▶ *Interactive Grid:View the complete power network visually.*
- ▶ *Failure Simulation:Apply failures to selected grid components.*
- ▶ *Live Monitoring:Track affected consumers and power allocation.*
- ▶ *Automatic Rerouting:Display alternate routes after a failure.*
- ▶ *Diagnostic Console:Show algorithm steps and simulation results.*

Integration of PBL Course Concepts

Course			Concepts Applied	Project Module
Data Structures in C++	OOPs with C++	in	Graphs, BFS, DFS, Dijkstra, Priority Queue, Min-Heap	Grid simulation and route finding
			Classes, Encapsulation, Inheritance, Polymorphism	Power stations, sub-stations and consumers
Operating Systems			Not directly used in the current project	N/A
DBMS				Grid/Graph data handling

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Major Components

- ▶ *Grid Backend* – Stores grid nodes and connections.
- ▶ *Algorithms* – BFS/DFS, Dijkstra, and Min-Heap handle routing and priority.
- ▶ *ImGui* – Displays the grid, failures, and results.

Data / Control Flow

- ▶ *User applies a failure to a grid component.*
- ▶ *BFS/DFS checks affected areas and grid connectivity.*
- ▶ *Dijkstra finds an alternate route and priority queue restores important loads.*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++17*
- ▶ Database: *std::unordered map*
- ▶ Development: *std::VS code*
- ▶ Tools / Framework: *Dear ImGui*
- ▶ IDE/Rendering: *GLFW + OpenGL*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. *Phase 1: Grid Initialization & Setup.*
2. *Phase 2: Phase 2: Normal Operation Testing.*
3. *Phase 3-4: Failure Simulation & Impact Analysis*
4. *Phase 5-7: Rerouting, Power Redistribution & Restoration*
5. *Phase 8: Results & Report Generation*

Justify the selection of technologies and methodology.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Grid setup and scenario certain completed.*
- ▶ *Graph structure and node connections implemented.*
- ▶ *Dear ImGui interface and control panel created.*
- ▶ *BFS, DFS and Dijkstra algorithms integrated.*
- ▶ *Priority-based power redistribution implemented.*

Individual Contribution

Member	Role	Contribution
Tejasv Singh Nagarkoti	System Design	Grid structure and overall project design
Tanishq Vashisth	Developer	Graph implementation and BFS/DFS
Tanishka Sahu	Developer	Dijkstra and power redistribution
Riddhiman Sandilya	GUI & Testing	GUI implementation and project testing

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: *Grid Setup & Initialization.*
2. Phase-II: *Failure Simulation & Routing.*
3. Phase-III: *Load Redistribution & Restoration.*

Expected Outcomes

- ▶ *Interactive visualization of power grid failures and recovery.*
- ▶ *Efficient rerouting and priority-based power redistribution.*

References

1. *GridRescue Project SOP Technical Specification.*
2. *Dear ImGui Documentation.*
3. *GLFW OpenGL Documentation.*
4. *Data Structures Graph Algorithm References.*

Thank You
Questions & Suggestions